

Confidence Layer

Optimising session quality, not conversion rate

Team Confidence Layer

FEG Innovation Hackathon 2026 · Challenge 1

Team Lead

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Solution architecture, Session
Quality Score design, demo
frontend.

Decisioning

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Architecture, Bandit loop,
hesitation classifier, reward
plumbing.

Data and Harm

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Harm-indicator models,
synthetic session generator,
holdout instrumentation.

The standard fix for hesitation is pressure.

In a regulated product, that fix is illegal.

When standard e-commerce platforms detect hesitation at the confirmation stage, they deploy a conventional playbook designed to force action. This playbook relies heavily on artificial urgency, such as countdown timers, aggressive social proof pop-ups, and sudden bonus inducements designed to override user deliberation.

However, in our context, these high-pressure tactics fundamentally fail FEG's strict responsible gambling guardrails. Furthermore, they are in direct violation of evolving Central and Eastern European iGaming compliance frameworks, which explicitly prohibit coercive design patterns.

Applying standard conversion rate optimization (CRO) techniques to a regulated environment introduces severe operational and legal risk. We cannot solve the hesitation problem by pressuring the user; we must find a mechanism that resolves their hesitation while strictly adhering to regulatory and ethical boundaries.

Hesitation at confirm is an information gap, not a motivation gap.

■ Read as pressure

Standard systems interpret hesitation as a lack of motivation, deploying urgency, aggressive social proof, and last-minute inducements.

This approach inevitably **raises harm indicators**, violates user trust, and completely fails the platform's responsible gambling guardrails. It forces a decision rather than facilitating one.

■ Read as information

Our system interprets hesitation as an information deficit. The user is pausing because they have an unresolved question: *'What does this market mean?'*, *'Did the odds move while I was browsing?'*, or *'Is this exactly what I picked?'*

By surfacing clarifying data at the exact moment of hesitation, we **resolve uncertainty without coercion**.

[TAKEAWAY] The bottom line is that FEG's brief explicitly asks for '**confident, informed**' actions. Providing clear, contextual information fulfills this mandate, whereas applying psychological pressure actively undermines it.

**Some abandonment is correct
behaviour the system must preserve.**

A hesitant user and a user about to make a poor decision look nearly identical in telemetry, and a system optimising conversion alone cannot represent the distinction at all.

We replace conversion rate with a score that has a minus sign in it.

To fundamentally alter the behavior of our optimization engine, we must change its mathematical objective. We introduce the Session Quality Score (SQS).

$$\text{SQS} = w_1 \cdot \text{action_value} + w_2 \cdot \text{discovery_efficiency} - w_3 \cdot \text{harm_indicator_load}$$

This formula balances positive engagement against protective constraints. The **action_value** rewards successful, intended actions, while **discovery_efficiency** rewards smooth navigation. Crucially, the **harm_indicator_load** is a subtractive term, heavily weighted and rendered in deep maroon (#8A2338) to signify the harm penalty.

Because the protection term is subtractive, any automated policy or intervention that inadvertently raises harm indicators cannot mathematically raise the overall score. This ensures that the reinforcement learning agent is mathematically prohibited from adopting predatory tactics to inflate short-term metrics.

Compliance stops being a rule we promise to follow and becomes a property of what we optimise.

01 By restructuring the reward function, we embed compliance directly into the system architecture. First, hard gates run before any scoring occurs to enforce deterministic compliance boundaries, ensuring that absolute rules are never subject to probabilistic overriding.

02 Second, the harm model stays architecturally independent of the conversion model, preventing the optimization engine from learning to suppress harm signals.

03 Third, because of this strict separation, a user refusal or a self-exclusion action cannot be optimized around by the system.

Visible Assumption: It is critical to acknowledge that this compliance guarantee holds true only while that architectural independence between the harm classifier and the reward function is strictly maintained in production.

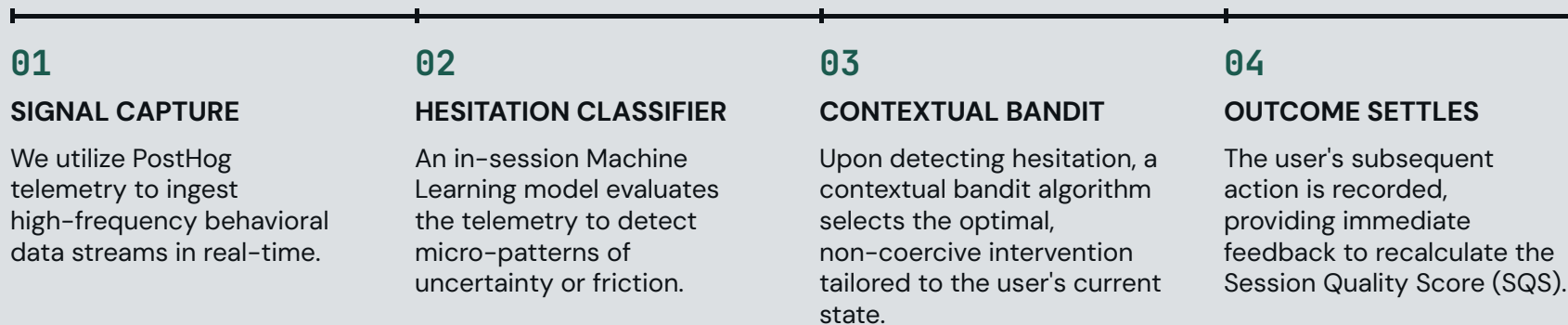
Detect, predict, intervene, settle— in-session, in under 100 milliseconds.

Our system operates continuously within the user's active session through a strict four-stage loop designed for extreme low-latency execution.

This entire cycle executes in under 100 milliseconds, ensuring the intervention feels seamless and native to the user experience.

<100ms

LOOP LATENCY



The intervention is a sentence of clarity, not a nudge.

When the system detects hesitation at the critical betslip confirmation moment, it does not deploy a flashing timer or a generic promotional push. Instead, it provides exact, contextual clarity.

"The odds moved since you added this; here is exactly what it returns. Keep it in your slip if you are not ready."

This approach directly addresses the information gap. Furthermore, we explicitly provide a low-friction defer path. The UI presents a clear, accessible button allowing the user to defer the action, save the slip for later, or set a reminder without any added friction.

By validating the user's hesitation and offering a safe exit, we build long-term trust and ensure that any completed action is genuinely confident and informed.

We fork the plumbing and write the scoring, because the scoring is what nobody has.

Our engineering strategy prioritizes velocity and domain-specific innovation by strictly separating commodity infrastructure from proprietary logic.

■ 01 / Fork (Infrastructure)

We leverage established, high-performance open-source tools for the underlying plumbing:

POSTHOG

Robust signal capture and event routing.

VOWPAL WABBIT / LEARN_TO_PICK

Contextual bandit mechanics referencing RLClientLib.

LIGHTGBM

Fast, efficient inference for the hesitation classifier.

■ 02 / Write (Proprietary Logic)

We dedicate custom engineering entirely to the layer that does not exist in the open-source ecosystem:

SESSION QUALITY SCORE (SQS)

Proprietary formula designed and written from scratch.

HARM INDICATOR PARSER

Highly specialized classification and detection engine.

ABSOLUTE COMPLIANCE CONTROL

Total control over responsible gambling design parameters.

We measure uplift against a stated baseline, with harm neutrality as a hard constraint.

Rigorously Defined Impact

We compare a simulated baseline against an optimized treatment group. If an intervention drives uplift but triggers any statistically significant increase in harm indicators, the system rejects the policy mathematically.

Minimal Client Footprint

- 01 One script tag for telemetry collection
- 02 One REST endpoint for real-time delivery

■ **+10.0%**

Optimized Treatment Uplift

■ **0.0%**

Harm-Indicator Tolerance

What ships in 24 hours, and what does not.

■ Team & Demo (24 Hours)

Our three-person team (**Lead Product Engineer, Machine Learning Specialist, Full-Stack Engineer, Responsible Gambling Policy Expert**) will deliver a functional prototype.

The live demo will run against simulated telemetry, successfully demonstrating the hesitation classifier, the contextual bandit decision loop, and the real-time calculation of the Session Quality Score.

■ Production Delta (Future Scope)

Moving this prototype to production requires building out a real-time PostHog pipeline capable of handling live scale, as well as establishing certified model guardrails to satisfy compliance auditors.

■ Risk Assessment

The single likeliest failure mode in a production environment is decision latency spikes exceeding the **100ms** threshold.

Our engineered fallback is to immediately default to a clean, intervention-free state, ensuring the user experience degrades gracefully without violating compliance.